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W-G-1115A
October 30, 1970
SUPERSEDING
Int. Fed. Spec.
W-G-001115(NAVY-Ships)
November 15, 1966
(See 6.8)

FEDERAL SPECIFICATION

GRINDER, ELECTRIC, PORTABLE, DOUBLE-INSULATED

(DIELECTRIC HOUSING)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 This specification covers one type of double-insulated, dielectric housing, portable electric grinder.

1.1.1 Federal Specification coverage. Federal specifications do not include all types, classes, styles and sizes of the commodity indicated by the title of this specification, but cover only those most generally used by the Federal Government.

1.2 Classification. The double-insulated, portable electric aerial grinder shall be of the horizontal type only and of the following size.

Size 6-6 inch diameter grinding wheel.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

Federal Specification:
PPP-P-40 - Packaging and Packing of Hand Tools.

Federal Standard:
Fed. Test Method Std. No. 406 - Plastics: Methods of Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications and Standards, and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:
MIL-V-173 - Varnish, Moisture and Fungus-Resistant (For the Treatment of Communications Electronic and Associated Electrical Equipment).

Military Standards:
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-1337 - Electric Portable Hand Tools General Suppression System Design Requirements.

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS
Handbook H-28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

UNDERWRITERS' LABORATORIES, INC. STANDARDS FOR SAFETY
UL 45 - Electric Tools.
UL 498 - Attachment Plugs and Recepticals.

(Application for copies should be addressed to Underwriters' Laboratories, Inc., 207 East Ohio Street, Chicago, Illinois 60611, 333 Pfingston Road, Northbrook, Illinois 60062, 1285 Walt Whitman Road, Melville, Long Island, New York 11746 and 1655 Scott Boulevard, Santa Clara, California 95113.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI STANDARDS)
B3.5 - Tolerances for Ball and Roller Bearings.
B3.6 - Boundary Dimensions for Ball and Roller Bearings.
B3.8 - Mounting Ball and Roller Bearings, Specifications For.
B5.38 - Driving and Spindle Ends for Portable Air and Electric Tools.
B7.1 - Safety Code for the Use, Care and Protection of Abrasive Wheels.
B74.2 - Shapes and Sizes of Grinding Wheels, Specifications for.

(Application for copies should be addressed to the American Standards Institute, 1430 Broadway, New York, New York 10018.)

THE ANTI-FRICTION BEARING MANUFACTURERS ASSOCIATION STANDARDS (AFBMA STANDARDS)
Section No. 5 - Bearing Identification Code for Anti-Friction Ball and Roller Bearings

(Application for copies should be addressed to the Anti-Friction Bearing Manufacturers Association, Inc., 60 East 42nd Street, New York, New York 10017.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Qualification. Double-insulated portable electric grinders furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.2 and 6.4).

3.2 Material. Materials used in fabrication of grinders shall be new, of uniform quality and be produced according to standard practices of reputable manufacturers producing equipment which will meet the requirements of this specification. Electrical materials and components shall conform to the Standards of the Underwriters' Laboratories (UL) as specified.

3.3 Insulation. Each tool shall be provided with double-insulation as defined (see 6.6.1). The purpose of a secondary protective insulation shall be to provide shock hazard protection in the event the primary insulation fails. Grinders shall conform to the insulation requirements of UL 45 except as specified herein.

3.4 Design. The grinders furnished shall be of the manufacturer's current design and shall include industry's latest developments in the double-insulated tool field. The grinders shall be of the commercial portable electric aerial grinder design. All parts subject to wear shall be easily accessible for adjustment, replacement, and repair.

3.4.1 Safety. Electric grinders shall be designed for ease of operation and safety in use. All rotating parts which are hazardous to the operator shall be suitably guarded against accidental contact where such guards do not interfere with direct operation of the tool. The enclosure, except for the ventilating openings, shall totally enclose all

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stationary and rotating electrical parts and provide adequate protection against accidental contact with live parts, even after the removal of detachable parts. Ventilating slots shall conform to the requirements of UL 45 and slot sizes shall be within the acceptable limit. Creepage or clearance distance through ventilating air slots from live parts to metal foil wrapped around the exterior of the case shall not be less than 3/16 inch. In addition to protection from moving parts, the grinders shall be so designed and constructed that in the event of defects which may occur in the functional insulation, they will cause no danger of shock to the operator.

3.4.1.1 Safety plate. In addition to the nameplate specified herein, a caution plate permanently marked with the following: "WARNING: USE ONLY GRINDING WHEELS THAT ARE MARKED SAFE AT OR ABOVE ACTUAL FREE RUNNING (NO-LOAD) SPEED (R.P.M.) OF TOOL," shall be attached to the grinder.

3.4.2 Creepage and clearance distance. The creepage and clearance distance along insulating surfaces and through air shall not be less than 3/32 inch between live parts and dead metal parts separated by primary functional insulation, between accessible dead metal parts and dead metal parts separated from live parts by primary insulation, between exterior of the housing and dead metal parts separated from live parts by functional insulation and 3/16 inch between live parts and accessible dead metal parts separated by reinforced insulation and between live parts and exterior housing of insulating material.

3.4.3 Lubrication. Bearings, except prelubricated sealed bearings, gears and moving parts shall be assembled with lubricating medium. Convenient access shall be provided to areas where manufacturer recommends periodic lubrication or lubricant replenishment is required. External lubricant fittings shall not be provided. Lubricant shall not leak from gear casing or enter motor compartment. Parts requiring protection from dust shall be enclosed.

3.4.4 Grinder intake air. Provision shall be made to reduce intake of dust and grinding debris through motor cooling air inlet openings.

3.4.5 Interchangeability. All parts from the same manufacturer shall be manufactured to such standards as will permit interchangeability of similar parts without modification or adjustment.

3.5 Construction. Construction shall be free of any characteristics or defects that will prevent the grinder from passing any of the examinations and tests specified in section 4. The grinder shall be complete so that when connected to the specified source of power, it can be used for any operation for which it was designed.

3.5.1 Fastening device. All screws, pins, bolts, and similar parts shall be installed with means for preventing loss of tightness. All such parts when subject to removal or adjustment shall not be swaged, peened, staked, or permanently deformed. All threaded parts shall be in the inch system and shall conform to Handbook H-28.

3.5.2 Gears and shafts. Gears and shafts shall be manufactured from hardened alloy steel. Gears and shafts shall be supported by at least two bearings and shall be of such dimensions to prevent deflection. Gears shall have machine cut or roll-formed teeth and be designed to efficiently transmit imposed loads. Gears and shafts, except end of spindle shaft, shall be fully enclosed.

3.5.3 Bearings. All rotating parts of the grinders shall be provided with precision quality commercially available ball and roller bearings conforming to B3.5 for tolerance, B3.6 for boundary dimensions, and B3.8 for mounting requirements and to AFBMA Standard Section No. 5 for identification. Integral or separate shielding to prevent leakage of lubricant and entrance of contaminants shall be provided where necessary. Steel inserts may be omitted in mounting of bearings when sufficient contact area and suitable fit between bearing outer-diameter and housing is provided.

3.5.3.1 Armature shaft bearings. Armature shaft bearings shall be ball or roller.

3.5.3.2 Spindle bearings. Spindle bearings shall be either ball, roller, or taper roller. If roller bearings are used, axial thrust loads applied to the spindle shaft must be provided for by use of a separate thrust bearing, either ball or roller or an integral combination type roller thrust bearing.

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3.6 Sizes and capacities. The grinder shall comply with the following capacities and dimensions as specified in table I.

Table I. Capacities and dimensions

Requirements	Size 6
Wheel Capacity, inches	
Diameter	6
Wheel face	1
Arbor hole	5/8
Spindle	
Diameter, inch, min.	5/8
Spindle thread	11 UNC
Wheel Speed, RPM	
No load, max.	4100
Full load, min.	2200
Horsepower, rated	
(Min.) at spindle	0.50 HP
Length, inches	
Overall, max.	28
Weight, pounds	
Net, max., complete with wheel and guard	25

3.7 Components. Grinders shall be of the materials specified and shall consist essentially of housing, handle, motor, switch, wiring, cord and plug, spindle, wheel guard, and abrasive grinding wheel.

3.7.1 Housings. External housings which are contacted by the operator shall be of self-extinguishing thermosetting plastic or equally insulative and dielectric material which will totally insulate the operator from the current-carrying parts of the tool. Metallic wheel guard and wear rings may be employed if totally isolated from electric shock. Internal metallic skeleton framework or shroud may be incorporated where required to meet the requirements specified herein. All exposed metal parts and attachments shall be completely insulated from current-carrying parts and internal metal parts by at least 0.080 inch of insulation by a creepage or clearance distance of at least 3/16 inch or an equal combination of the two. Housings shall include the motor case housing, the spindle housing, and handle and switch housing. Spindle and handle housings shall be designed as convenient hand grips.

3.7.1.1 Molded parts of housing. Flash shall be removed from molded parts. Flash edges shall be smooth. To prevent excessive moisture absorption into the volume of parts made of thermosetting or thermoplastic material, buffed or cut areas may be given two coats of varnish conforming to MIL-V-173 if necessary for the grinder to meet the 168-hour humidity test in accordance with 4.5.6.2.

3.7.2 Handle. Grinders shall have straight type handle at the end opposite the grinding wheel. Handle shall be of an insulating material, shall house the switch and be designed to provide a convenient grip for the operator. A comfortable gripping surface shall also be provided on the spindle housing for two-hand operation.

3.7.3 Motor. The motor shall be universal series wound or compensated series wound designed to operate on either 115 volts (±6 percent) single phase 25 to 60 cycles alternating current or 115 volts direct current. Motors shall conform to the requirements of UL 45.

3.7.3.1 Rotor. The rotor (armature) shall be balanced dynamically to insure freedom from distructive vibration.

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3.7.3.2 Motor shaft insulation. Secondary or protecting insulation (double-insulation) shall be provided between the armature shaft and the lamination core and commutator.

3.7.3.3 Commutators. Commutators shall be either of the "V" ring and sheet mica or the mica molded construction. Molded commutators shall be of material that will withstand maximum rotational forces and be immune to deformation from temperatures encountered in service.

3.7.3.4 Brushes. The motor shall be furnished with brushes fitted in suitable insulated brush holders, and shall be provided with suitable devices to exert the proper operating pressure upon the brushes. The minimum internal creepage shall be 3/16 inch, measured from the edge of the live brush holder over the insulation to the adjacent metal part which can be touched from the outside of the housing. Where insulated screw caps are used, they shall be protected by the housing from mechanical injury. Brushes and brush holders shall be such that:

(a) Brush will not rotate in holder.
(b) No live parts are exposed.
(c) Adequate brush pressure is maintained throughout life of brush.
(d) Brush springs shall not be affected by operating temperature.
(e) Brushes can be easily replaced indefinitely without deforming any part of the brush holder.

3.7.4 Switch. The grinder shall be provided with a switch of the quick break design for turning the motor "off" and "on." The switch shall be encased in high dielectric insulating material and shall comply with UL 45 requirements for switches used in double-insulated tools. The switch shall release automatically and shall have a lock for holding in the "on" position. The switch shall have a current rating of not less than the rated load rating of the grinder and shall be capable of interrupting stalled motor current at rated voltage for 50 consecutive contacts at 10-second intervals without sticking or burning of the contacts.

3.7.5 Wiring, cord, and plug. All wiring, and flexible cord, shall be in accordance with UL 45 as applicable. The cord shall have an attachment plug cap and unless otherwise specified (see 6.2) shall have a minimum free length of not less than 9 feet measured from the point of entry into the grinder. The attachment plug cap shall comply with UL 498 requirements for two-pole, two wire devices rated at 15 amperes, 125 volts.

3.7.6 Spindle. Spindle shall be alloy steel, hardened and finished to the diameter specified in table I. Spindle shall be mounted on ball or roller bearing with integral or separate shielding and sealed to prevent leakage of lubricant or entrance of contaminants. Spindle shall be insulated from current-carrying parts by protecting insulation as well as the functional insulation. Spindle end threads and dimensions shall conform to B5.38 for straight wheel grinders.

3.7.7 Wheel guard. Wheel guard shall conform to the requirements of B7.1.

3.7.8 Grinding wheel. Each grinder shall be furnished with a grinding wheel of the dimensions specified in table I. Wheel shall be square face, 24 to 36 grit, vitrified or resinoid bond and shall conform to type I shape of B74.2. The grinding wheel shall be mounted on the grinder clamped between two flange washers of identical outside diameter. Washers (blotters) shall be used between wheel and each flange. Flange washers and blotters shall conform to B7.1 for the type and size wheels provided.

3.8 Fungus proofing. When specified (see 6.2), the electric circuitry, including all components and connections, shall be protected from the effect of water and fungus growth by an overall treatment with a varnish conforming to MIL-V-173 with the following exceptions:

(a) Components or circuit elements which are inherently fungus and water resistant or which are hermetically sealed, need not be treated.
(b) Components or circuit elements whose function will be adversely affected by the varnish coating shall not be treated.

The varnish, when used, shall be applied to give a minimum dry thickness of 0.001 inch to component or element surface previously cleaned and prepared in accordance with accepted commercial practice.

3.9 Electromagnetic interference suppression. When specified (see 6.2), grinders shall be electromagnetic interference suppressed within the limits specified for portable electric equipment in compliance with the requirements of MIL-STD-1337.

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3.10 Nameplate. Each grinder shall be provided with a suitable nameplate which shall be permanently marked and securely fastened to an integral part of the frame of the housing in a conspicuous location. The nameplate shall give the manufacturer's name, type or model number, serial number, size or capacity of the grinder, type of current (a.c./d.c.), voltage, rated-load current in amperes, no load r.p.m. and the words "DOUBLE-INSULATED" or symbol denoting double-insulation shall be included on nameplate or in a prominent position on grinder. If a metallic nameplate is used, it shall be secured in such a way that it will not interfere with the double-insulating feature of the grinder. Creepage distance requirements as specified shall be complied with. Grinders shall have one of the following notations on the nameplate:

"This tool requires two (2) prong plug only."
"Grounding not required."

3.11 Instruction book. The manufacturer shall furnish one copy of his regular instruction book and parts list with each grinder. The instruction book shall include all illustrations, instructions, and other necessary data for safety operation, maintenance, adjustment and repair of the tool. The parts list shall include views of part identified by manufacturer's part number, and ordering instructions.

3.12 Lifting eye. When specified (see 6.2), each grinder shall be provided with a lifting eye for aerial suspension. The lifting eye shall be positioned for balance.

3.13 Extra parts and accessories. In addition to the accessories specified herein, each grinder shall be furnished with such maintenance tools and extra parts as normally furnished commercially.

3.14 Workmanship. Workmanship of the grinders shall be of the quality prevailing among manufacturers normally producing grinders of the double-insulated type and sizes as specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Qualification tests.[1/] Qualification tests shall be conducted in a laboratory satisfactory to the Naval Ship Engineering Center. Qualification tests shall consist of all the examinations and tests required for quality conformance and the additional tests of 4.5.3, 4.5.4, 4.5.5, 4.5.6, and 4.5.7. Failure of the grinder in any test shall result in termination of the qualification tests.

4.2.1 Samples for qualification tests. The manufacturer shall submit one grinder complete with instruction book, parts list, and two extra sets of brushes for qualification tests.

4.2.2 Prior approval of electromagnetic interference suppression. Prior to authorization for qualification, complete electromagnetic interference tests as specified herein shall be performed by the contractor or a commercial test laboratory to determine compliance with the electromagnetic suppression requirements. Three copies of the interference suppression test report shall be furnished to the Naval Ship Engineering Center for review and approval.

4.3 Sampling for quality conformance.

4.3.1 Lot. All double-insulated grinders offered for delivery at one time shall be considered a lot for purposes of quality conformance inspection.

4.3.2 Sample. A random sample shall be selected from each lot in conformance with MIL-STD-105 at inspection level II.

[1/] Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6.4 and 6.5).

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4.4 Quality conformance inspection. Quality conformance inspection shall be performed on each sample grinder selected in accordance with 4.3.2. Quality conformance inspection shall consist of the following:

(a) Examination (4.4.1).
(b) Tests (4.4.2).
(c) Inspection of preservation and packaging (4.6).

4.4.1 Examination. Each grinder selected in accordance with 4.3.2 shall be examined to determine compliance with the requirements of this specification. Examination shall be conducted in accordance with table II, acceptable quality level to be as indicated.

Table II. Classification of defects

#	Defects
Critical:	AQL 0.10 percent defective
001	Not manufacturer's model designation as listed on Qualified Products List.
Major:	AQL 2.5 percent defective
101	Size not as specified.
102	Nameplate and safety plate nonconforming.
103	Handle and spindle hand hold not comfortable or suitable for 2-hand control.
104	Switch not type specified.
105	Locking device for switch not included.
106	Cord and plug not as specified.
107	Wheel guard missing or not in accord with applicable standard.
108	Accessories not as specified in order.
109	Workmanship not of quality expected of good commercial manufacturing procedures.
110	Spindle size and threads not as specified.
Minor:	AQL 4.0 percent defective
201	Instruction books not as specified.

4.4.2 Tests for quality conformance. Each sample selected in accordance with 4.3.2 shall be tested for compliance with the requirements of this specification. AQL shall be 2.5 percent defective. Nonconforming grinders shall not be offered for delivery and if the number of defective units exceeds the acceptable number for the samples tested, the lot represented shall be rejected. Quality conformance tests shall consist of the tests specified in 4.5.1, 4.5.2 and 4.5.3.

4.5 Tests.

4.5.1 General. The electromagnetic interference suppression system shall be disconnected during the tests of 4.5.5 and 4.5.6.

4.5.1.1 Electromagnetic interference test report. When specified (see 6.2), manufacturer shall provide certification that the electromagnetic suppression network in the grinder offered is identical to that of grinders previously tested and approved prior to qualification.

4.5.2 Dielectric strength electrical circuits. Each sample grinder shall be tested for electric circuit dielectric strength by withstanding 4000 volts RMS, 60 cycle, alternating current from line to exposed metal surface with switch "on" for a period of one second.

4.5.3 Operational test. Sample grinders shall be operated 10 minutes at no-load operation. After the no-load operational period, rated load conditions (rated current, voltage and frequency in conformance with UL 45 requirements) shall apply. The grinder shall be operated at rated load for a period of 30 minutes. During the test the grinder shall be checked for switch operation and loss of lubricant. Leakage current shall be determined and shall not exceed 1 milliampere. Minimum insulation resistance of at least 2 megohms shall be determined at power leads (500 volts for a period of 1 minute). Nonconforming grinders shall be rejected.

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4.5.4 Performance test. Sample grinders shall be tested for the requirements specified herein and as listed on nameplate:

(a) Amperes at rated load on 115 volts, 60 cycles, AC current.
(b) Horsepower output at spindle.
(c) Spindle speed.
    (1) No load.
    (2) Rated load.

4.5.5 Protective insulation tests. The protective insulation tests shall be conducted in the following order: 4.5.5.1, 4.5.5.2, 4.5.5.3, 4.5.5.4, 4.5.5.5, and 4.5.5.6. The inside of the grinder housing and the grinder internals shall not be cleaned in any way during this test sequence.

4.5.5.1 Flammability test. At least three sample specimens, 1/2 inch wide, 5 inches long by the material thickness, shall be cut from a sheet or molded part of the identical protective insulation used in fabrication of the tool or from tool test sample housings. Sample specimens shall conform to the requirements and test procedures of Fed. Test Method Std. No. 406 Method 2021.

4.5.5.2 Humidity test. Before any other tests are run, the grinder shall be placed in a humidity chamber for a period of one week (approximately 168 hours) at 95 percent relative humidity and 25° to 30°C. At the end of the humidification period and within 5 minutes after removal from the humidity chamber, the insulation resistance and leakage current shall be measured and recorded. The insulation resistance shall not be less than 1 megohm and the leakage current shall not exceed 5 milliamperes.

4.5.5.2.1 Solution used in humidity tests. Humidity tests (4.5.5.2 and 4.5.5.5) shall be prepared using a saturated solution of Potassium Sulphate (K_2SO_4) dissolved in distilled water. Undissolved crystals shall be visible in solution container at all times. 95±3 percent R.H. shall be controlled by maintaining cabinet temperature between 20° and 30°C.

4.5.5.3 No load running and brush wear test. The grinder shall be run continuously at no load (with normal air flow through grinder) for 100 hours. A balanced grinding wheel or simulated metallic wheel of equal weight shall be installed during test. Failure of any component during this test shall be cause for rejection of the grinder.

4.5.5.4 Leakage current test. The tool shall be running at no load on 110 percent of the rated voltage marked on the unit. The leakage current shall then be measured from each power supply conductor to accessible metal parts and to metal foil applied to the exterior surface of the protective insulated housing. The leakage current shall not exceed 3.5 milliamperes (0.0035 amperes) from either power supply conductor, either with or without electromagnetic interference filters installed.

4.5.5.5 Insulation resistance test. Grinder shall be placed in a humidity cabinet for 48 hours. Temperature in cabinet shall be 20° to 30°C. and a relative humidity of between 90 and 95 percent shall be maintained. Upon removal from cabinet the insulation resistance shall be measured using an applied potential of 500 volts DC between live parts via plug and switch and accessible dead metal parts and between live parts and metal foil applied to the exterior surface of the protective insulated housing or case. Minimum insulation shall be 1 megohm. Nonconforming grinders shall be rejected.

4.5.5.6 Insulation dielectric tests. A dielectric test shall be performed immediately after the insulation resistance test. Sixty cycle potential shall be applied for at least one minute without breakdown at the following voltages: (All voltages are RMS values unless otherwise stated.)

(a) Between live metal parts and inaccessible dead metal parts 1000 volts plus 2 times rated voltage.
(b) Between inaccessible dead metal parts and accessible dead metal parts and also between inacessible dead metal parts and metal foil applied to exterior surface of grinder 2000 volts plus 2 times rated voltage.
(c) Between metal foil applied to interior of compartments containing internal wiring and accessible dead metal parts or metal foil attached to exterior surface of the grinder 2000 volts plus 2 times rated voltage.
(d) Between metal foil wrapped around power cord at the junction with tool and metal foil attached to exterior of grinder or accessible dead metal parts 2000 volts plus 2 times rated voltage.
(e) Between live parts and accessible dead metal parts and between live parts and metal foil applied to exterior of grinder 3500 volts plus 2 times rated voltage.

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4.5.6 Impact tests. The grinder shall be tested as follows for compliance with the impact resistance qualities of the protective insulating material. The grinder shall be dropped (without wheel) at least 3 times on a hard wood surface from a distance of 3 feet in such a position that the grinder will strike a different position on each drop.

After the above tests, the grinder shall be examined and will be considered as having failed the tests if there is:

(a) Evidence of cracking case or housing.
(b) Reduction of spacing below minimum required or exposure to live parts.
(c) Evidence that internal parts have been damaged or grinder fails to operate due to broken insulation or internal parts.

4.5.7 Insulation stability test. (This is a destructive test to be conducted last for the purpose of determining safety properties of insulation.) The grinder shall be connected to rated voltage and frequency and loaded using a dynamometer, prony brake or other loading device. The current input shall be 1.8 times the rated input marked on the tool. This load shall be maintained for one hour or until the motor fails. Following this overload, the following requirements shall be met:

(a) There will be no exposed live parts.
(b) Spacings will not be less than the minimum required by this specification.
(c) A potential of 2,500 volts 60 cycle shall be applied for one minute without breakdown between live parts and accessible dead metal parts and between live parts and metal foil applied to the exterior surface of housing made of insulating material.

4.6 Inspection of preparation for delivery requirements. An inspection shall be made to determine that the preservation, packaging, packing, and marking comply with the requirements in section 5 of this specification. Defects shall be scored in accordance with table III. For examination of interior packaging the sample unit shall be one shipping container fully prepared for delivery, selected at random just prior to the closing operations. Sampling shall be in accordance with MIL-STD-105. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 with an acceptable quality level (AQL) of 4.0 defects per hundred units.

Table III. Classification of preparation for delivery defects

Examine	Defects
Markings (exterior and interior)	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Materials	Any component missing, or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strappings, or distortion of container.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. Grinders shall be preserved, packaged, packed, and marked in accordance with PPP-P-40. The level of preservation and packaging and the level of packing shall be as specified (see 6.2).

6. NOTES

6.1 Intended use. The double-insulated grinder as furnished under this specification is hand guided, hand pressure applied, portable electric aerial, for use in foundries, forge shops, welding shops, and general grinding operations in the removal of burrs, fins, flash, excess weld material, sprues, and similar hand grinding operations. When mounted on a stand, it can be used as a bench grinder. The grinder is particularly recommended for use aboard ships or in damp locations and outside work because of its double-insulated design.

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6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Cord length, if different (see 3.7.5).
(c) Moisture and fungus resistance treatment, when required (see 3.8).
(d) Electromagnetic interference suppression when required (see 3.9 and 4.5.1.1).
(e) When grinder is to be provided with lifting eye (see 3.12).
(f) Level of preservation and packaging and level of packing required (see 5.1).

6.3 Supersession data. This specification includes the requirements of Military Specification MIL-G-24089A(SHIPS) dated 15 July 1965.

6.4 With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in applicable Qualified Products List QPL W-G-1115, whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Ship Engineering Center, Center Building, Prince George's Center, Hyattsville, Maryland 20782, and information pertaining to qualification of products may be obtained from that activity. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6.5).

6.5 Copies of "Provisions Governing Qualification SD-6" may be obtained upon application to Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

6.6 Definitions. Definitions are as follows:

6.6.1 Double-insulation. Insulation comprising both functional insulation and protecting insulation, with the two insulations physically separated and so arranged that they are not simultaneously subjected to the same deteriorating influences (temperatures and contaminants) to the same degree.

6.6.1.1 Functional insulation. Insulation necessary for the proper functioning of the tool and for basic protection against electric shock.

6.6.1.2 Protecting insulation. Denotes an independent insulation provided, in addition to the functional insulation, in order to ensure protection against electric shock in case of failure of the functional insulation. In addition, enclosures (housings) of insulation material shall form a part or the whole of the protecting insulation.

6.6.1.3 Reinforced insulation. Denotes an improved functional insulation with such mechanical and electrical qualities that it provides the same degree of protection against electric shock as double-insulation. The enclosures of insulating material may form a part or the whole of the reinforced insulation.

6.7 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in Section 2 do not apply when material and parts are procured by the supplier for incorporation into the equipment and lose their separate identity when the equipment is shipped.

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6.8 CHANGES FROM PREVIOUS ISSUE. THE OUTSIDE MARGINS OF THIS DOCUMENT HAVE BEEN MARKED "#" TO INDICATE WHERE CHANGES (DELETIONS, ADDITIONS, ETC.) FROM THE PREVIOUS ISSUE HAVE BEEN MADE. THIS HAS BEEN DONE AS A CONVENIENCE ONLY AND THE GOVERNMENT ASSUMES NO LIABILITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIONS. BIDDERS AND CONTRACTORS ARE CAUTIONED TO EVALUATE THE REQUIREMENTS OF THIS DOCUMENT BASED ON THE ENTIRE CONTENT AS WRITTEN IRRESPECTIVE OF THE MARGINAL NOTATIONS AND RELATIONSHIP TO THE LAST PREVIOUS ISSUE.

MILITARY INTEREST:

Custodians:

Army - GL
Navy - SH
Air Force - 84

Review activities:

Army - EL, GL
Navy - SH
Air Force - 84

User activities:

Army - ME
Navy - AS, OS, CG, YD

Preparing activity:

Navy - SH

CIVIL AGENCY INTEREST:

GSA-FSS

Project No. 5130-0134

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price 15 cents each.

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